

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T46 · Congruence, Similarity and Geometrical Proof

With Answers

Questions with full worked solutions

■ 16 pages

FAMILY MAP

Congruence Similarity Geometrical Proof

F01

Use length scale factor in similar figures

7 Q | 19 marks

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

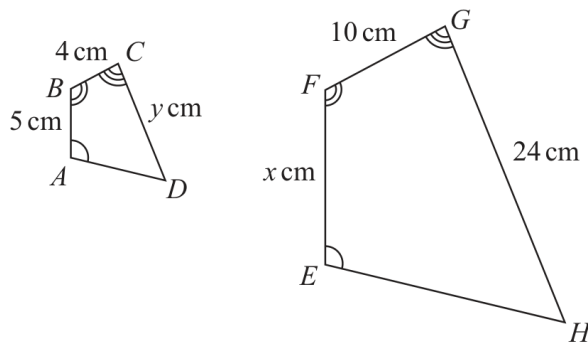


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

- (a) Work out the value of x

$$x = \dots\dots\dots (2)$$

Continued
4MA1-2H Q5

ORIGINAL QUESTION

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

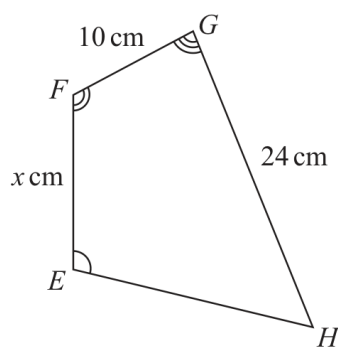
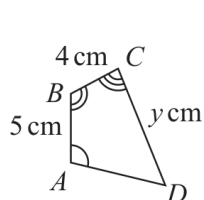


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

- (b) Work out the value of y

$$y = \dots\dots\dots (2)$$

Worked Solution 20

Similar quadrilaterals

UNIT 2**4 marks****Scale factor**

$$\frac{FG}{BC} = \frac{10}{4} = 2.5.$$

Find x and y

$$x = 5(2.5) = 12.5, \quad y = \frac{24}{2.5} = 9.6.$$

Final answer

$$x = 12.5 \text{ cm}, \quad y = 9.6 \text{ cm}$$

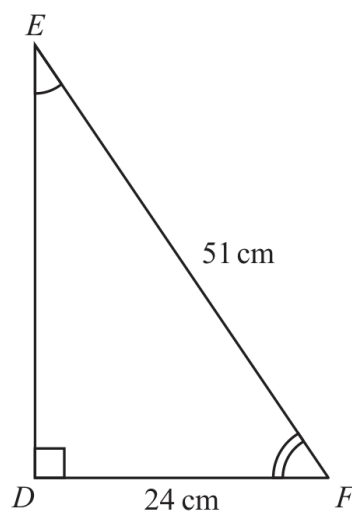
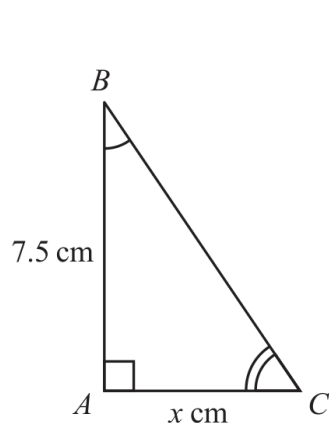
Original question 10

4MA1-1H Q10 M01

MODULAR UNIT 2

5 marks

10 Here are two similar triangles.



Diagrams **NOT**
accurately drawn

Work out the value of x
Show your working clearly.

$x =$

Worked solution 10Use similarity to find x

MODULAR UNIT 2

5 marks

M01 – Use similarity to find x **Find the long corresponding side**

$$DE^2 = 51^2 - 24^2 = 2025 \Rightarrow DE = 45.$$

Find the scale factor

$$\text{small:large} = 7.5/45 = 1/6.$$

Scale the corresponding side

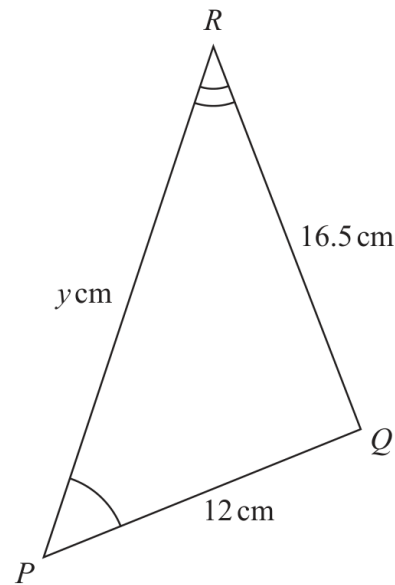
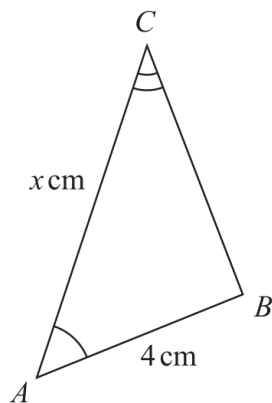
$$x = 24(1/6) = 4 \text{ cm.}$$

Final answer

4 cm

Original question 16

4MA1-2H Q3 Q3(a)

ORIGINAL QUESTION**2 marks****3**Diagram **NOT**
accurately drawnTriangle ABC is similar to triangle PQR

$$AB = 4 \text{ cm} \quad PQ = 12 \text{ cm} \quad RQ = 16.5 \text{ cm} \quad AC = x \text{ cm} \quad PR = y \text{ cm}$$

(a) Calculate the length of BC

..... cm
(2)

Worked solution 16

Chapter: Congruence, Similarity & Geometrical Proof

MODULAR UNIT 2

2 marks

Q3(a) – Chapter: Congruence, Similarity & Geometrical Proof

Family: Use length scale factor in similar figures • Variant: Find a corresponding length in similar triangles

MethodThe scale factor from triangle ABC to PQR is $PQ/AB=12/4=3$.*Why: Corresponding lengths in similar triangles have a constant ratio.***Method** $BC=16.5/3=5.5$ cm.*Why: Divide the corresponding length by the scale factor.***Final answer****BC: 5.5 cm**

Original question 17

4MA1-2H Q3 Q3(b)

ORIGINAL QUESTION**1 marks**

(b) Write down an expression for y in terms of x

$y = \dots\dots\dots$
(1)

Worked solution 17

Chapter: Congruence, Similarity & Geometrical Proof

MODULAR UNIT 2

1 marks

Q3(b) – Chapter: Congruence, Similarity & Geometrical Proof

Family: Use length scale factor in similar figures • Variant: Solve a chained or algebraic similarity ratio

MethodThe corresponding side PR is three times AC, so $y=3x$.*Why: Use the same similarity scale factor.***Final answer**

$$y: y=3x$$

4 ABC and DEF are similar triangles.

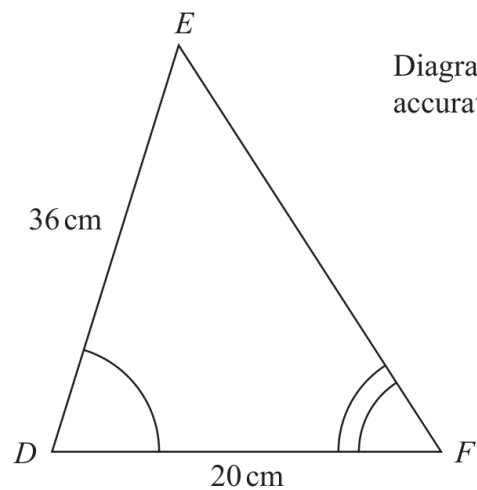
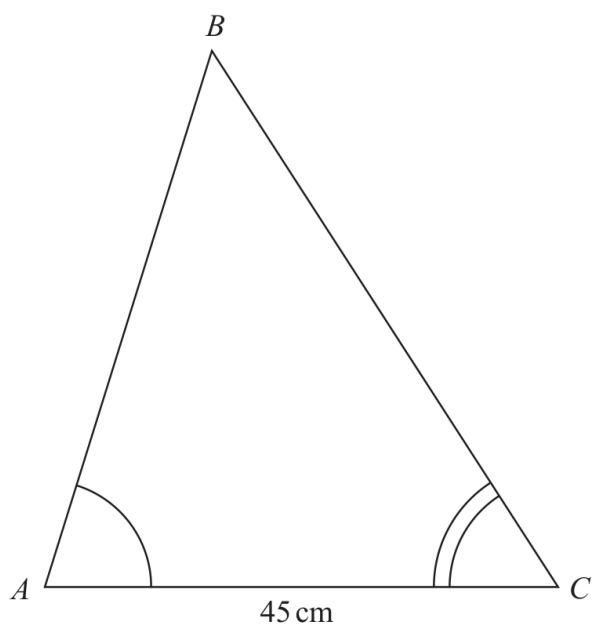


Diagram **NOT**
accurately drawn

(a) Work out the length of AB .

..... cm
(2)

Worked solution

November 2021 | Question 4 | 2 marks

Family: Use length scale factor in similar figures | Variant: Find a corresponding length in similar triangles

METHOD / WORKING

1. The scale factor from triangle DEF to triangle ABC is

2. $\frac{AC}{DF} = \frac{45}{20} = 2.25$

3. $AB = DE \times 2.25 = 36 \times 2.25 = 81$

4. Given $BC = 54$,

5. $EF = \frac{54}{2.25} = 24$

6. Answer: $AB = 81\text{cm}$, and $EF = 24\text{cm}$.

FINAL ANSWER

AB=81 cm, and EF=24 cm.

Given that $BC = 54\text{ cm}$,

(b) work out the length of EF .

..... cm
(2)

Worked solution

November 2021 | Question 4 | 2 marks

Family: Use length scale factor in similar figures | Variant: Find a corresponding length in similar triangles

METHOD / WORKING

1. The scale factor from triangle DEF to triangle ABC is

2. $\frac{AC}{DF} = \frac{45}{20} = 2.25$

3. $AB = DE \times 2.25 = 36 \times 2.25 = 81$

4. Given $BC = 54$,

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6. Answer: $AB = 81\text{cm}$, and $EF = 24\text{cm}$.

FINAL ANSWER

AB=81 cm, and EF=24 cm.

4 Here are three similar quadrilaterals.

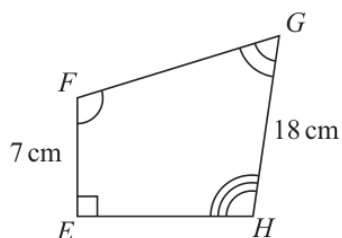
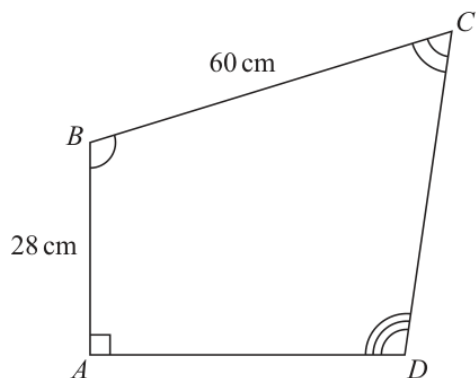
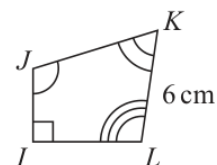


Diagram **NOT** accurately drawn



Work out the length of JK

..... cm

(Total for Question 4 is 3 marks)

Worked solution

November 2025 | Question 4 | 3 marks

Family: Use length scale factor in similar figures • Variant: Find corresponding lengths in similar polygons

Solution

The scale factor from the largest quadrilateral to the middle quadrilateral is

$$(7)/(28)=\text{frac}14$$

So the side corresponding to 60 cm in the middle shape is

$$60 \times \text{frac}14=15$$

The scale factor from the middle quadrilateral to the smallest quadrilateral is

$$(6)/(18)=\text{frac}13$$

$$JK=15 \times \text{frac}13=5$$

Answer: 5 cm.